```
> restart;with(linalg);assume(a>0);assume(e<1,e>0);assume(ep<1,
  ep>0);
  assume(ap>0);
  assume(omega,real);assume(Omega,real);assume(i>0,i<Pi);
  assume(v,real);assume(u,real);assume(GM>0);assume(beta>0,beta<1);
  Digits:=20;with(plots);
```

[*BlockDiagonal*, *GramSchmidt*, *JordanBlock*, *LUdecomp*, *QRdecomp*, *Wronskian*, *addcol*, *addrow*, *adj*, *adjoint*, *angle*, *augment*, *backsub*, *band*, *basis*, *bezout*, *blockmatrix*, *charmat*, *charpoly*, *cholesky*, *col*, *coldim*, *colspace*, *colspan*, *companion*, *concat*, *cond*, *copyinto*, *crossprod*, *curl*, *definite*, *delcols*, *delrows*, *det*, *diag*, *diverge*, *dotprod*, *eigenvals*, *eigenvalues*, *eigenvectors*, *eigenvects*, *entermatrix*, *equal*, *exponential*, *extend*, *ffgausselim*, *fibonacci*, *forwardsub*, *frobenius*, *gausselim*, *gaussjord*, *geneqns*, *genmatrix*, *grad*, *hadamard*, *hermite*, *hessian*, *hilbert*, *htranspose*, *ihermite*, *indexfunc*, *innerprod*, *intbasis*, *inverse*, *ismith*, *issimilar*, *iszero*, *jacobian*, *jordan*, *kernel*, *laplacian*, *leastsqrs*, *linsolve*, *matadd*, *matrix*, *minor*, *minpoly*, *mulcol*, *mulrow*, *multiply*, *norm*, *normalize*, *nullspace*, *orthog*, *permanent*, *pivot*, *potential*, *randmatrix*, *randvector*, *rank*, *ratform*, *row*, *rowdim*, *rowspace*, *rowspan*, *rref*, *scalarmul*, *singularvals*, *smith*, *stackmatrix*, *submatrix*, *subvector*, *sumbasis*, *swapcol*, *swaprow*, *sylvester*, *toeplitz*, *trace*, *transpose*, *vandermonde*, *vecpotent*, *vectdim*, *vector*, *wronskian*]

$$Digits := 20$$

[*animate*, *animate3d*, *animatecurve*, *arrow*, *changecoords*, *complexplot*, *complexplot3d*, *conformal*, *conformal3d*, *contourplot*, *contourplot3d*, *coordplot*, *coordplot3d*, *densityplot*, *display*, *dualaxisplot*, *fieldplot*, *fieldplot3d*, *gradplot*, *gradplot3d*, *implicitplot*, *implicitplot3d*, *inequal*, *interactive*, *interactiveparams*, *intersectplot*, *listcontplot*, *listcontplot3d*, *listdensityplot*, *listplot*, *listplot3d*, *loglogplot*, *logplot*, *matrixplot*, *multiple*, *odeplot*, *pareto*, *plotcompare*, *pointplot*, *pointplot3d*, *polarplot*, *polygonplot*, *polygonplot3d*, *polyhedra_supported*, *polyhedraplot*, *rootlocus*, *semilogplot*, *setcolors*, *setoptions*, *setoptions3d*, *shadebetween*, *spacecurve*, *sparsematrixplot*, *surfdata*, *textplot*, *textplot3d*, *tubeplot*] **(1)**

Base

```
> cO:=cos(Omega);sO:=sin(Omega);so:=sin(omega);co:=cos(omega);
  ci:=cos(i);si:=sin(i);
```

$$cO := \cos(\Omega\sim)$$
$$sO := \sin(\Omega\sim)$$
$$so := \sin(\omega\sim)$$
$$co := \cos(\omega\sim)$$
$$ci := \cos(i\sim)$$
$$si := \sin(i\sim) \quad \textbf{(1.1)}$$

```
> E1:=vector([co*cO-so*ci*sO,co*sO+so*ci*cO,so*si]);
  E2:=vector([-so*cO-co*ci*sO,-so*sO+co*cO*ci,co*si]);
```

$$E1 := [-\cos(i\sim)\sin(\Omega\sim)\sin(\omega\sim) + \cos(\Omega\sim)\cos(\omega\sim), \cos(\Omega\sim)\cos(i\sim)\sin(\omega\sim) + \cos(\omega\sim)\sin(\Omega\sim), \sin(\omega\sim)\sin(i\sim)]$$
$$E2 := [-\cos(i\sim)\cos(\omega\sim)\sin(\Omega\sim) - \cos(\Omega\sim)\sin(\omega\sim), \cos(\Omega\sim)\cos(i\sim)\cos(\omega\sim) - \sin(\Omega\sim)\sin(\omega\sim), \cos(\omega\sim)\sin(i\sim)] \quad (1.2)$$

```
> n:=sqrt(GM)/a^(3/2);rr:=a*(1-e*cos(u));rx:=a*(cos(u)-e);
  ry:=a*sqrt(1-e^2)*sin(u);vx:=-n*a^2/rr*sin(u);
  vy:=n*a^2/rr*sqrt(1-e^2)*cos(u);
  #cv:=(cos(u)-e)/(1-e*cos(u));sv:=sqrt(1-e^2)*sin(u)/(1-e*cos(u)
  );
```

$$n := \frac{\sqrt{GM\sim}}{a\sim^{3/2}}$$
$$rr := a\sim(1 - e\sim\cos(u\sim))$$
$$rx := a\sim(\cos(u\sim) - e\sim)$$
$$ry := a\sim\sqrt{-e\sim^2 + 1}\,\sin(u\sim)$$
$$vx := -\frac{\sqrt{GM\sim}\,\sin(u\sim)}{\sqrt{a\sim}\,(1 - e\sim\cos(u\sim))}$$
$$vy := \frac{\sqrt{GM\sim}\,\sqrt{-e\sim^2 + 1}\,\cos(u\sim)}{\sqrt{a\sim}\,(1 - e\sim\cos(u\sim))} \quad (1.3)$$

```
> vtou:={cos(v)=(cos(u)-e)/(1-e*cos(u)),
         sin(v)=sqrt(1-e^2)*sin(u)/(1-e*cos(u))};
```

$$vtou := \left\{\cos(v\sim) = \frac{\cos(u\sim) - e\sim}{1 - e\sim\cos(u\sim)}, \sin(v\sim) = \frac{\sqrt{-e\sim^2 + 1}\,\sin(u\sim)}{1 - e\sim\cos(u\sim)}\right\} \quad (1.4)$$

```
> utov:={sin(u)=sqrt(1-e^2)*sin(v)/(1+e*cos(v)),
  cos(u)=(e+cos(v))/(1+e*cos(v))};
```

$$utov := \left\{\cos(u\sim) = \frac{e\sim + \cos(v\sim)}{1 + e\sim\cos(v\sim)}, \sin(u\sim) = \frac{\sqrt{-e\sim^2 + 1}\,\sin(v\sim)}{1 + e\sim\cos(v\sim)}\right\} \quad (1.5)$$

$$> exq := \mathrm{sqrt}(1 - e^2);\ dvdM := \frac{(1 + e\cdot\cos(v))^2}{exq^3};\ sqgl := \mathrm{sqrt}(2\cdot(1 - exq));$$
$$dvdMu := factor(simplify(subs(vtou, dvdM)));$$
$$exq := \sqrt{-e\sim^2 + 1}$$
$$dvdM := \frac{(1 + e\sim\cos(v\sim))^2}{(-e\sim^2 + 1)^{3/2}}$$
$$sqgl := \sqrt{2 - 2\sqrt{-e\sim^2 + 1}}$$
$$dvdMu := \frac{\sqrt{-(e\sim - 1)(e\sim + 1)}}{(e\sim\cos(u\sim) - 1)^2} \quad (1.6)$$

```
> pos:=evalm(rx*E1+ry*E2);
```

$$pos := \Big[a\sim\,(\cos(u\sim) - e\sim)\,(-\cos(i\sim)\sin(\Omega\sim)\sin(\omega\sim) + \cos(\Omega\sim)\cos(\omega\sim)) + a\sim\sqrt{-e\sim^2+1}\,\sin(u\sim)\,(-\cos(i\sim)\cos(\omega\sim)\sin(\Omega\sim) - \cos(\Omega\sim)\sin(\omega\sim)),\ a\sim\,(\cos(u\sim) - e\sim)\,(\cos(\Omega\sim)\cos(i\sim)\sin(\omega\sim) + \cos(\omega\sim)\sin(\Omega\sim)) + a\sim\sqrt{-e\sim^2+1}\,\sin(u\sim)\,(\cos(\Omega\sim)\cos(i\sim)\cos(\omega\sim) - \sin(\Omega\sim)\sin(\omega\sim)),\ a\sim\,(\cos(u\sim) - e\sim)\sin(\omega\sim)\sin(i\sim) + a\sim\sqrt{-e\sim^2+1}\,\sin(u\sim)\cos(\omega\sim)\sin(i\sim)\Big] \quad (1.7)$$

```
> vit:=evalm(vx*E1+vy*E2);
```

$$vit := \Bigg[-\frac{\sqrt{GM\sim}\,\sin(u\sim)\,(-\cos(i\sim)\sin(\Omega\sim)\sin(\omega\sim) + \cos(\Omega\sim)\cos(\omega\sim))}{\sqrt{a\sim}\,(1 - e\sim\cos(u\sim))} + \frac{1}{\sqrt{a\sim}\,(1 - e\sim\cos(u\sim))}\Big(\sqrt{GM\sim}\,\sqrt{-e\sim^2+1}\,\cos(u\sim)\,(-\cos(i\sim)\cos(\omega\sim)\sin(\Omega\sim) - \cos(\Omega\sim)\sin(\omega\sim))\Big),$$

$$-\frac{\sqrt{GM\sim}\,\sin(u\sim)\,(\cos(\Omega\sim)\cos(i\sim)\sin(\omega\sim) + \cos(\omega\sim)\sin(\Omega\sim))}{\sqrt{a\sim}\,(1 - e\sim\cos(u\sim))} + \frac{1}{\sqrt{a\sim}\,(1 - e\sim\cos(u\sim))}\Big(\sqrt{GM\sim}\,\sqrt{-e\sim^2+1}\,\cos(u\sim)\,(\cos(\Omega\sim)\cos(i\sim)\cos(\omega\sim) - \sin(\Omega\sim)\sin(\omega\sim))\Big),$$

$$-\frac{\sqrt{GM\sim}\,\sin(u\sim)\sin(\omega\sim)\sin(i\sim)}{\sqrt{a\sim}\,(1 - e\sim\cos(u\sim))} + \frac{\sqrt{GM\sim}\,\sqrt{-e\sim^2+1}\,\cos(u\sim)\cos(\omega\sim)\sin(i\sim)}{\sqrt{a\sim}\,(1 - e\sim\cos(u\sim))}\Bigg] \quad (1.8)$$

```
> ;
```

Cas 1 (astrométrie, masse libre)

$$> f1 := \frac{e\cdot\cos(\Omega+\omega)}{exq};\ f2 := \frac{e\cdot\sin(\Omega+\omega)}{exq};$$

$$f3 := \mathrm{sqrt}(exq)\cdot\sin\left(\frac{i}{2}\right)\cdot\cos(\mathrm{omega}-\Omega);\ f4 := \mathrm{sqrt}(exq)\cdot\sin\left(\frac{i}{2}\right)\cdot\sin(\mathrm{omega}$$

$-\Omega);$

$f5 := \frac{\cos(\Omega + \omega + \nu)}{P};$

$f6 := \frac{\sin(\Omega + \omega + \nu)}{P};$

$f7 := \ln(a);$

$$f1 := \frac{e\sim \cos(\Omega\sim + \omega\sim)}{\sqrt{-e\sim^2 + 1}}$$

$$f2 := \frac{e\sim \sin(\Omega\sim + \omega\sim)}{\sqrt{-e\sim^2 + 1}}$$

$$f3 := (-e\sim^2 + 1)^{1/4} \sin\left(\frac{1}{2}\, i\sim\right) \cos(-\omega\sim + \Omega\sim)$$

$$f4 := -(-e\sim^2 + 1)^{1/4} \sin\left(\frac{1}{2}\, i\sim\right) \sin(-\omega\sim + \Omega\sim)$$

$$f5 := \frac{\cos(\Omega\sim + \omega\sim + \nu\sim)}{P}$$

$$f6 := \frac{\sin(\Omega\sim + \omega\sim + \nu\sim)}{P}$$

$$f7 := \ln(a\sim) \tag{2.1}$$

```
> A := matrix([[diff(f1, e), diff(f1, ω), diff(f1, i), diff(f1, Ω), diff(f1, P),
    diff(f1, ν), diff(f1, a)],
        [diff(f2, e), diff(f2, ω), diff(f2, i), diff(f2, Ω), diff(f2, P), diff(f2,
    ν), diff(f2, a)],
        [diff(f3, e), diff(f3, ω), diff(f3, i), diff(f3, Ω), diff(f3, P), diff(f3,
    ν), diff(f3, a)],
        [diff(f4, e), diff(f4, ω), diff(f4, i), diff(f4, Ω), diff(f4, P), diff(f4,
    ν), diff(f4, a)],
        [diff(f5, e), diff(f5, ω), diff(f5, i), diff(f5, Ω), diff(f5, P), diff(f5,
    ν), diff(f5, a)],
        [diff(f6, e), diff(f6, ω), diff(f6, i), diff(f6, Ω), diff(f6, P), diff(f6,
    ν), diff(f6, a)],
        [diff(f7, e), diff(f7, omega), diff(f7, i), diff(f7, Omega), diff(f7, P),
    diff(f7, ν), diff(f7, a)]]);
  ;
```

$$A := \left[\left[\frac{\cos(\Omega\sim + \omega\sim)}{\sqrt{-e\sim^2 + 1}} + \frac{e\sim^2 \cos(\Omega\sim + \omega\sim)}{(-e\sim^2 + 1)^{3/2}}, -\frac{e\sim \sin(\Omega\sim + \omega\sim)}{\sqrt{-e\sim^2 + 1}}, 0, -\frac{e\sim \sin(\Omega\sim + \omega\sim)}{\sqrt{-e\sim^2 + 1}}, 0, 0, 0\right], \tag{2.2}$$

$$\left[\frac{\sin(\Omega\sim+\omega\sim)}{\sqrt{-e\sim^2+1}}+\frac{e\sim^2\sin(\Omega\sim+\omega\sim)}{(-e\sim^2+1)^{3/2}},\frac{e\sim\cos(\Omega\sim+\omega\sim)}{\sqrt{-e\sim^2+1}},0,\frac{e\sim\cos(\Omega\sim+\omega\sim)}{\sqrt{-e\sim^2+1}},0,0,0\right],$$

$$\left[-\frac{1}{2}\frac{\sin\left(\frac{1}{2}i\sim\right)\cos(-\omega\sim+\Omega\sim)\,e\sim}{(-e\sim^2+1)^{3/4}},(-e\sim^2+1)^{1/4}\sin\left(\frac{1}{2}i\sim\right)\sin(-\omega\sim+\Omega\sim),\frac{1}{2}(-e\sim^2+1)^{1/4}\cos\left(\frac{1}{2}i\sim\right)\cos(-\omega\sim+\Omega\sim),-(-e\sim^2+1)^{1/4}\sin\left(\frac{1}{2}i\sim\right)\sin(-\omega\sim+\Omega\sim),0,0,0\right],$$

$$\left[\frac{1}{2}\frac{\sin\left(\frac{1}{2}i\sim\right)\sin(-\omega\sim+\Omega\sim)\,e\sim}{(-e\sim^2+1)^{3/4}},(-e\sim^2+1)^{1/4}\sin\left(\frac{1}{2}i\sim\right)\cos(-\omega\sim+\Omega\sim),-\frac{1}{2}(-e\sim^2+1)^{1/4}\cos\left(\frac{1}{2}i\sim\right)\sin(-\omega\sim+\Omega\sim),-(-e\sim^2+1)^{1/4}\sin\left(\frac{1}{2}i\sim\right)\cos(-\omega\sim+\Omega\sim),0,0,0\right],$$

$$\left[0,-\frac{\sin(\Omega\sim+\omega\sim+\nu\sim)}{P},0,-\frac{\sin(\Omega\sim+\omega\sim+\nu\sim)}{P},-\frac{\cos(\Omega\sim+\omega\sim+\nu\sim)}{P^2},-\frac{\sin(\Omega\sim+\omega\sim+\nu\sim)}{P},0\right],$$

$$\left[0,\frac{\cos(\Omega\sim+\omega\sim+\nu\sim)}{P},0,\frac{\cos(\Omega\sim+\omega\sim+\nu\sim)}{P},-\frac{\sin(\Omega\sim+\omega\sim+\nu\sim)}{P^2},\frac{\cos(\Omega\sim+\omega\sim+\nu\sim)}{P},0\right],$$

$$\left[0,0,0,0,0,0,\frac{1}{a\sim}\right]\Bigg]$$

> $simplify(linalg[det](A)), factor\left(simplify\left(\frac{linalg[det](A)\cdot P\cdot a}{2\cdot\sin\left(\frac{i}{2}\right)\cdot\cos\left(\frac{i}{2}\right)}\cdot dvdM\right)\right),$
> $factor\left(simplify\left(\frac{linalg[det](A)\cdot P\cdot a}{2\cdot\sin\left(\frac{i}{2}\right)\cdot\cos\left(\frac{i}{2}\right)}\cdot dvdMu\right)\right);$

On multiplie par P et a et on divise par sin(i) pour tenir compte des priors (ln P, ln a, et sin(i)) sur P, a et i

p(pars).d(pars)=C.d(pars)=C.J.d(elts)=C.J.d(e).d(omega).d(i).d(Omega).d(P).d(nu).d

(a)
=C.J.d(e).d(omega).d(Omega).d(cos(i)).d(lnP).d(lna).d(M).P.a.(d(nu)/dM)/sin(i)

$$\frac{e\sim\cos\left(\frac{1}{2}\,i\sim\right)\sin\left(\frac{1}{2}\,i\sim\right)}{\left(-e\sim^2+1\right)^{3/2}P^3\,a\sim},\ -\frac{1}{2}\,\frac{e\sim\,(1+e\sim\cos(\nu\sim))^2}{(e\sim-1)^3\,(e\sim+1)^3\,P^2},\ -\frac{1}{2}\,\frac{e\sim}{(e\sim-1)\,(e\sim+1)\,P^2\,(e\sim\cos(u\sim)-1)^2} \tag{2.3}$$

>

Cas 1bis : Vitesses radiales seules, masse contrainte

> $f1 := \frac{e\cdot\cos(\omega)}{exq}$; $f2 := \frac{e\cdot\sin(\omega)}{exq}$;
$f3 := \frac{\cos(\omega+\nu)}{P}$;
$f4 := \frac{\sin(\omega+\nu)}{P}$;
$f5 := \ln(m)$;

$$f1 := \frac{e\sim\cos(\omega\sim)}{\sqrt{-e\sim^2+1}}$$
$$f2 := \frac{e\sim\sin(\omega\sim)}{\sqrt{-e\sim^2+1}}$$
$$f3 := \frac{\cos(\omega\sim+\nu\sim)}{P}$$
$$f4 := \frac{\sin(\omega\sim+\nu\sim)}{P}$$
$$f5 := \ln(m) \tag{3.1}$$

> $A := matrix([[diff(f1, e), diff(f1, \omega), diff(f1, P), diff(f1, \nu), diff(f1, m)],$
$[diff(f2, e), diff(f2, \omega), diff(f2, P), diff(f2, \nu), diff(f2, m)],$
$[diff(f3, e), diff(f3, \omega), diff(f3, P), diff(f3, \nu), diff(f3, m)],$
$[diff(f4, e), diff(f4, \omega), diff(f4, P), diff(f4, \nu), diff(f4, m)],$
$[diff(f5, e), diff(f5, \omega), diff(f5, P), diff(f5, \nu), diff(f5, m)]]);$

$$A := \left[\left[\frac{\cos(\omega\sim)}{\sqrt{-e\sim^2+1}}+\frac{e\sim^2\cos(\omega\sim)}{\left(-e\sim^2+1\right)^{3/2}},\ -\frac{e\sim\sin(\omega\sim)}{\sqrt{-e\sim^2+1}},0,0,0\right],\right.$$
$$\left[\frac{\sin(\omega\sim)}{\sqrt{-e\sim^2+1}}+\frac{e\sim^2\sin(\omega\sim)}{\left(-e\sim^2+1\right)^{3/2}},\ \frac{e\sim\cos(\omega\sim)}{\sqrt{-e\sim^2+1}},0,0,0\right],$$
$$\left[0,\ -\frac{\sin(\omega\sim+\nu\sim)}{P},\ -\frac{\cos(\omega\sim+\nu\sim)}{P^2},\ -\frac{\sin(\omega\sim+\nu\sim)}{P},0\right], \tag{3.2}$$

$$\left[0, \frac{\cos(\omega\sim + \nu\sim)}{P}, -\frac{\sin(\omega\sim + \nu\sim)}{P^2}, \frac{\cos(\omega\sim + \nu\sim)}{P}, 0\right],$$
$$\left[0, 0, 0, 0, \frac{1}{m}\right]\Big]$$

> $simplify(linalg[det](A)), factor(simplify(linalg[det](A)\cdot P\cdot m\cdot dvdM)),$
$factor(simplify(linalg[det](A)\cdot P\cdot m\cdot dvdMu));$

On multiplie par P et m pour tenir compte
des priors (ln P, ln m) sur P, a
p(pars).d(pars)=C.d(pars)=C.J.d(elts)=C.J.d(e).d(omega).d(Omega).d(P).d(nu).d(m)
=C.J.d(e).d(omega).d(Omega).d(lnP).d(lnm).d(M).P.a.(d(nu)/dM)

$$-\frac{e\sim}{(e\sim^2-1)^2 P^3 m}, -\frac{e\sim(1+e\sim\cos(\nu\sim))^2}{P^2(-(e\sim-1)(e\sim+1))^{7/2}}, -\frac{e\sim}{(-(e\sim-1)(e\sim+1))^{3/2} P^2 (e\sim\cos(u\sim)-1)^2} \quad \textbf{(3.3)}$$

>

Cas 2 : - masse libre

> $f1 := \frac{e\cdot\cos(\text{Omega}+\text{omega})}{exq}; f2 := \frac{e\cdot\sin(\text{Omega}+\text{omega})}{exq};$
$f3 := e^{\text{xi}}\cdot\text{sqrt}(exq)\cdot\sin\left(\frac{i}{2}\right)\cdot\cos(\text{omega}-\text{Omega}); f4 := e^{\text{xi}}\cdot\text{sqrt}(exq)\cdot\sin\left(\frac{i}{2}\right)$
$\cdot\sin(\text{omega}-\text{Omega});$
$f5 := \frac{1}{P}\cdot\cos(\text{Omega}+\text{omega}+\nu);$
$f6 := \frac{1}{P}\cdot\sin(\text{Omega}+\text{omega}+\nu);$
$f7 := \ln(a);$

$$f1 := \frac{e\sim\cos(\Omega\sim+\omega\sim)}{\sqrt{-e\sim^2+1}}$$
$$f2 := \frac{e\sim\sin(\Omega\sim+\omega\sim)}{\sqrt{-e\sim^2+1}}$$
$$f3 := e\sim^{\xi}(-e\sim^2+1)^{1/4}\sin\left(\frac{1}{2}i\sim\right)\cos(-\omega\sim+\Omega\sim)$$
$$f4 := -e\sim^{\xi}(-e\sim^2+1)^{1/4}\sin\left(\frac{1}{2}i\sim\right)\sin(-\omega\sim+\Omega\sim)$$
$$f5 := \frac{\cos(\Omega\sim+\omega\sim+\nu\sim)}{P}$$
$$f6 := \frac{\sin(\Omega\sim+\omega\sim+\nu\sim)}{P}$$

$$f7 := \ln(a\sim) \tag{4.1}$$

```
> A := matrix([[diff(f1, e), diff(f1, ω), diff(f1, i), diff(f1, Ω), diff(f1, P),
    diff(f1, ν), diff(f1, a)],
        [diff(f2, e), diff(f2, ω), diff(f2, i), diff(f2, Ω), diff(f2, P), diff(f2,
    ν), diff(f2, a)],
        [diff(f3, e), diff(f3, ω), diff(f3, i), diff(f3, Ω), diff(f3, P), diff(f3,
    ν), diff(f3, a)],
        [diff(f4, e), diff(f4, ω), diff(f4, i), diff(f4, Ω), diff(f4, P), diff(f4,
    ν), diff(f4, a)],
        [diff(f5, e), diff(f5, ω), diff(f5, i), diff(f5, Ω), diff(f5, P), diff(f5,
    ν), diff(f5, a)],
        [diff(f6, e), diff(f6, ω), diff(f6, i), diff(f6, Ω), diff(f6, P), diff(f6,
    ν), diff(f6, a)],
        [diff(f7, e), diff(f7, omega), diff(f7, i), diff(f7, Omega), diff(f7, P),
    diff(f7, ν), diff(f7, a)]]);
;
```

$$A := \left[\left[\frac{\cos(\Omega\sim+\omega\sim)}{\sqrt{-e\sim^2+1}}+\frac{e\sim^2\cos(\Omega\sim+\omega\sim)}{(-e\sim^2+1)^{3/2}}, -\frac{e\sim\sin(\Omega\sim+\omega\sim)}{\sqrt{-e\sim^2+1}}, 0, -\frac{e\sim\sin(\Omega\sim+\omega\sim)}{\sqrt{-e\sim^2+1}}, 0, 0, 0\right],\right. \tag{4.2}$$

$$\left[\frac{\sin(\Omega\sim+\omega\sim)}{\sqrt{-e\sim^2+1}}+\frac{e\sim^2\sin(\Omega\sim+\omega\sim)}{(-e\sim^2+1)^{3/2}}, \frac{e\sim\cos(\Omega\sim+\omega\sim)}{\sqrt{-e\sim^2+1}}, 0, \frac{e\sim\cos(\Omega\sim+\omega\sim)}{\sqrt{-e\sim^2+1}}, 0, 0, 0\right],$$

$$\left[\frac{e\sim^{\xi}\xi(-e\sim^2+1)^{1/4}\sin\left(\frac{1}{2}i\sim\right)\cos(-\omega\sim+\Omega\sim)}{e\sim}\right.$$

$$-\frac{1}{2}\frac{e\sim^{\xi}\sin\left(\frac{1}{2}i\sim\right)\cos(-\omega\sim+\Omega\sim)\,e\sim}{(-e\sim^2+1)^{3/4}}, e\sim^{\xi}(-e\sim^2+1)^{1}$$

$$^{/4}\sin\left(\frac{1}{2}i\sim\right)\sin(-\omega\sim+\Omega\sim), \frac{1}{2}e\sim^{\xi}(-e\sim^2+1)^{1/4}\cos\left(\frac{1}{2}i\sim\right)\cos(-\omega\sim$$

$$+\Omega\sim), -e\sim^{\xi}(-e\sim^2+1)^{1/4}\sin\left(\frac{1}{2}i\sim\right)\sin(-\omega\sim+\Omega\sim), 0, 0, 0\Big],$$

$$\left[-\frac{e\sim^{\xi}\xi(-e\sim^2+1)^{1/4}\sin\left(\frac{1}{2}i\sim\right)\sin(-\omega\sim+\Omega\sim)}{e\sim}\right.$$

$$+\frac{1}{2}\,\frac{e\sim^{\xi}\sin\left(\frac{1}{2}\,i\sim\right)\sin(-\omega\sim+\Omega\sim)\,e\sim}{(-e\sim^{2}+1)^{3/4}},\ e\sim^{\xi}(-e\sim^{2}+1)^{1/4}\sin\left(\frac{1}{2}\,i\sim\right)\cos(-\omega\sim+\Omega\sim),\ -\frac{1}{2}\,e\sim^{\xi}(-e\sim^{2}+1)^{1/4}\cos\left(\frac{1}{2}\,i\sim\right)\sin(-\omega\sim+\Omega\sim),\ -e\sim^{\xi}(-e\sim^{2}+1)^{1/4}\sin\left(\frac{1}{2}\,i\sim\right)\cos(-\omega\sim+\Omega\sim),\ 0,\ 0,\ 0\Big],$$

$$\Big[0,\ -\frac{\sin(\Omega\sim+\omega\sim+\nu\sim)}{P},\ 0,\ -\frac{\sin(\Omega\sim+\omega\sim+\nu\sim)}{P},\ -\frac{\cos(\Omega\sim+\omega\sim+\nu\sim)}{P^{2}},\ -\frac{\sin(\Omega\sim+\omega\sim+\nu\sim)}{P},\ 0\Big],$$

$$\Big[0,\ \frac{\cos(\Omega\sim+\omega\sim+\nu\sim)}{P},\ 0,\ \frac{\cos(\Omega\sim+\omega\sim+\nu\sim)}{P},\ -\frac{\sin(\Omega\sim+\omega\sim+\nu\sim)}{P^{2}},\ \frac{\cos(\Omega\sim+\omega\sim+\nu\sim)}{P},\ 0\Big],$$

$$\Big[0,\ 0,\ 0,\ 0,\ 0,\ 0,\ \frac{1}{a\sim}\Big]\Big]$$

> $simplify(linalg[det](A)),\ factor\left(simplify\left(\frac{linalg[det](A)\cdot P\cdot a}{2\cdot\sin\left(\frac{i}{2}\right)\cdot\cos\left(\frac{i}{2}\right)}\cdot dvdM\right)\right);$

$$factor\left(simplify\left(\frac{linalg[det](A)\cdot P\cdot a}{2\cdot\sin\left(\frac{i}{2}\right)\cdot\cos\left(\frac{i}{2}\right)}\cdot dvdMu\right)\right)$$

On multiplie par P et a et on divise par cos(i) pour tenir compte des priors (ln a, ln P et sin(i))
sur a, P et i (voir plus haut)

$$\frac{e\sim^{1+2\xi}\cos\left(\frac{1}{2}\,i\sim\right)\sin\left(\frac{1}{2}\,i\sim\right)}{(-e\sim^{2}+1)^{3/2}\,P^{3}\,a\sim},\ -\frac{1}{2}\,\frac{e\sim^{1+2\xi}(1+e\sim\cos(\nu\sim))^{2}}{(e\sim-1)^{3}\,(e\sim+1)^{3}\,P^{2}}-\frac{1}{2}\,\frac{e\sim^{1+2\xi}}{(e\sim-1)\,(e\sim+1)\,P^{2}\,(e\sim\cos(u\sim)-1)^{2}} \tag{4.3}$$

>

Cas 2bis : - masse libre, e=0, omega=0

> $f1 := \sin\left(\frac{i}{2}\right)\cdot\cos(\mathrm{Omega});\ f2 := -\sin\left(\frac{i}{2}\right)\cdot\sin(\mathrm{Omega});$

$f3 := \frac{1}{P}\cdot\cos(\mathrm{Omega}+\nu);$

$f4 := \frac{1}{P} \cdot \sin(\text{Omega} + v);$

$f5 := \ln(a);$

$$f1 := \sin\left(\frac{1}{2}\, i\sim\right) \cos(\Omega\sim)$$

$$f2 := -\sin\left(\frac{1}{2}\, i\sim\right) \sin(\Omega\sim)$$

$$f3 := \frac{\cos(\Omega\sim + v\sim)}{P}$$

$$f4 := \frac{\sin(\Omega\sim + v\sim)}{P}$$

$$f5 := \ln(a\sim) \tag{5.1}$$

```
> A := matrix([[diff(f1, i), diff(f1, Ω), diff(f1, P), diff(f1, v), diff(f1, a)],
              [diff(f2, i), diff(f2, Ω), diff(f2, P), diff(f2, v), diff(f2, a)],
              [diff(f3, i), diff(f3, Ω), diff(f3, P), diff(f3, v), diff(f3, a)],
              [diff(f4, i), diff(f4, Ω), diff(f4, P), diff(f4, v), diff(f4, a)],
              [diff(f5, i), diff(f5, Ω), diff(f5, P), diff(f5, v), diff(f5, a)]]);
  ;
```

$$A := \left[\left[\frac{1}{2} \cos\left(\frac{1}{2}\, i\sim\right) \cos(\Omega\sim), -\sin\left(\frac{1}{2}\, i\sim\right) \sin(\Omega\sim), 0, 0, 0\right], \left[-\frac{1}{2} \cos\left(\frac{1}{2}\, i\sim\right) \sin(\Omega\sim), -\sin\left(\frac{1}{2}\, i\sim\right) \cos(\Omega\sim), 0, 0, 0\right], \left[0, -\frac{\sin(\Omega\sim + v\sim)}{P}, -\frac{\cos(\Omega\sim + v\sim)}{P^2}, -\frac{\sin(\Omega\sim + v\sim)}{P}, 0\right], \left[0, \frac{\cos(\Omega\sim + v\sim)}{P}, -\frac{\sin(\Omega\sim + v\sim)}{P^2}, \frac{\cos(\Omega\sim + v\sim)}{P}, 0\right], \left[0, 0, 0, 0, \frac{1}{a\sim}\right]\right] \tag{5.2}$$

```
> simplify(linalg[det](A)), factor(simplify(subs(e = 0, linalg[det](A)·P·a / (2·sin(i/2)·cos(i/2)) ·dvdM))), factor(simplify(subs(e = 0, linalg[det](A)·P·a / (2·sin(i/2)·cos(i/2)) ·dvdMu)));
```

On multiplie par P et a et on divise par cos(i) pour tenir compte des priors (ln a, ln P et sin(i))
sur a, P et i (voir plus haut)

$$\frac{1}{2} \frac{\cos\left(\frac{1}{2}\, i\sim\right) \sin\left(\frac{1}{2}\, i\sim\right)}{P^3\, a\sim}, \frac{1}{4\, P^2}, \frac{1}{4\, P^2} \tag{5.3}$$

```
>
```

Cas 3 : Astrométrie, masse contrainte

> $g1 := \frac{e \cdot \cos(\Omega + \omega)}{exq}; g2 := \frac{e \cdot \sin(\Omega + \omega)}{exq};$
$g3 := e^{xi} \cdot \sin\left(\frac{i}{2}\right) \cdot \cos(\text{omega-}\Omega) \cdot \text{sqrt}(exq); g4 := e^{xi} \cdot \text{sqrt}(exq) \cdot \sin\left(\frac{i}{2}\right) \cdot \sin(\text{omega-}\Omega);$
$g5 := \frac{1}{P} \cdot \cos(\Omega + \omega + \nu);$
$g6 := \frac{1}{P} \cdot \sin(\Omega + \omega + \nu);$

$$g1 := \frac{e\sim \cos(\Omega\sim + \omega\sim)}{\sqrt{-e\sim^2 + 1}}$$
$$g2 := \frac{e\sim \sin(\Omega\sim + \omega\sim)}{\sqrt{-e\sim^2 + 1}}$$
$$g3 := e\sim^{\xi} \sin\left(\frac{1}{2}\, i\sim\right) \cos(-\omega\sim + \Omega\sim)\, (-e\sim^2 + 1)^{1/4}$$
$$g4 := -e\sim^{\xi} (-e\sim^2 + 1)^{1/4} \sin\left(\frac{1}{2}\, i\sim\right) \sin(-\omega\sim + \Omega\sim)$$
$$g5 := \frac{\cos(\Omega\sim + \omega\sim + \nu\sim)}{P}$$
$$g6 := \frac{\sin(\Omega\sim + \omega\sim + \nu\sim)}{P} \tag{6.1}$$

> $A := matrix([[diff(g1, e), diff(g1, \omega), diff(g1, i), diff(g1, \Omega), diff(g1, P), diff(g1, \nu)],$
$[diff(g2, e), diff(g2, \omega), diff(g2, i), diff(g2, \Omega), diff(g2, P), diff(g2, \nu)],$
$[diff(g3, e), diff(g3, \omega), diff(g3, i), diff(g3, \Omega), diff(g3, P), diff(g3, \nu)],$
$[diff(g4, e), diff(g4, \omega), diff(g4, i), diff(g4, \Omega), diff(g4, P), diff(g4, \nu)],$
$[diff(g5, e), diff(g5, \omega), diff(g5, i), diff(g5, \Omega), diff(g5, P), diff(g5, \nu)],$
$[diff(g6, e), diff(g6, \omega), diff(g6, i), diff(g6, \Omega), diff(g6, P), diff(g6, \nu)]]);$

$$A := \left[\left[\frac{\cos(\Omega\sim + \omega\sim)}{\sqrt{-e\sim^2 + 1}} + \frac{e\sim^2 \cos(\Omega\sim + \omega\sim)}{(-e\sim^2 + 1)^{3/2}}, -\frac{e\sim \sin(\Omega\sim + \omega\sim)}{\sqrt{-e\sim^2 + 1}}, 0, -\frac{e\sim \sin(\Omega\sim + \omega\sim)}{\sqrt{-e\sim^2 + 1}}, 0, 0\right], \right. \tag{6.2}$$

$$\left[\frac{\sin(\Omega\sim+\omega\sim)}{\sqrt{-e\sim^2+1}}+\frac{e\sim^2\sin(\Omega\sim+\omega\sim)}{(-e\sim^2+1)^{3/2}},\frac{e\sim\cos(\Omega\sim+\omega\sim)}{\sqrt{-e\sim^2+1}},0,\frac{e\sim\cos(\Omega\sim+\omega\sim)}{\sqrt{-e\sim^2+1}},0,0\right],$$

$$\left[\frac{e\sim^{\xi}\xi\sin\left(\frac{1}{2}i\sim\right)\cos(-\omega\sim+\Omega\sim)(-e\sim^2+1)^{1/4}}{e\sim}-\frac{1}{2}\frac{e\sim^{\xi}\sin\left(\frac{1}{2}i\sim\right)\cos(-\omega\sim+\Omega\sim)e\sim}{(-e\sim^2+1)^{3/4}},e\sim^{\xi}(-e\sim^2+1)^{1/4}\sin\left(\frac{1}{2}i\sim\right)\sin(-\omega\sim+\Omega\sim),\frac{1}{2}e\sim^{\xi}\cos\left(\frac{1}{2}i\sim\right)\cos(-\omega\sim+\Omega\sim)(-e\sim^2+1)^{1/4},-e\sim^{\xi}(-e\sim^2+1)^{1/4}\sin\left(\frac{1}{2}i\sim\right)\sin(-\omega\sim+\Omega\sim),0,0\right],$$

$$\left[-\frac{e\sim^{\xi}\xi(-e\sim^2+1)^{1/4}\sin\left(\frac{1}{2}i\sim\right)\sin(-\omega\sim+\Omega\sim)}{e\sim}+\frac{1}{2}\frac{e\sim^{\xi}\sin\left(\frac{1}{2}i\sim\right)\sin(-\omega\sim+\Omega\sim)e\sim}{(-e\sim^2+1)^{3/4}},e\sim^{\xi}\sin\left(\frac{1}{2}i\sim\right)\cos(-\omega\sim+\Omega\sim)(-e\sim^2+1)^{1/4},-\frac{1}{2}e\sim^{\xi}(-e\sim^2+1)^{1/4}\cos\left(\frac{1}{2}i\sim\right)\sin(-\omega\sim+\Omega\sim),-e\sim^{\xi}\sin\left(\frac{1}{2}i\sim\right)\cos(-\omega\sim+\Omega\sim)(-e\sim^2+1)^{1/4},0,0\right],$$

$$\left[0,-\frac{\sin(\Omega\sim+\omega\sim+\nu\sim)}{P},0,-\frac{\sin(\Omega\sim+\omega\sim+\nu\sim)}{P},-\frac{\cos(\Omega\sim+\omega\sim+\nu\sim)}{P^2},-\frac{\sin(\Omega\sim+\omega\sim+\nu\sim)}{P}\right],$$

$$\left[0,\frac{\cos(\Omega\sim+\omega\sim+\nu\sim)}{P},0,\frac{\cos(\Omega\sim+\omega\sim+\nu\sim)}{P},-\frac{\sin(\Omega\sim+\omega\sim+\nu\sim)}{P^2},\frac{\cos(\Omega\sim+\omega\sim+\nu\sim)}{P}\right]\Bigg]$$

> $simplify(linalg[det](A)), factor\left(simplify\left(\frac{linalg[det](A)\cdot P}{2\cdot\sin\left(\frac{i}{2}\right)\cdot\cos\left(\frac{i}{2}\right)}\cdot dvdM\right)\right);$

(6.3)

$$\frac{e\sim^{1+2\xi}\cos\left(\frac{1}{2}\,i\sim\right)\sin\left(\frac{1}{2}\,i\sim\right)}{\left(-e\sim^{2}+1\right)^{3/2}P^{3}},\ -\frac{1}{2}\,\frac{e\sim^{1+2\xi}\left(1+e\sim\cos(\nu\sim)\right)^{2}}{\left(e\sim-1\right)^{3}\left(e\sim+1\right)^{3}P^{2}} \tag{6.3}$$

```
> -147.5 + 180;
```

$$32.5 \tag{6.4}$$

Cas 5 : Astrométrie+vitesses radiales, mass contrainte

```
> f1 := e·cos(ω)/exq; f2 := e·sin(ω)/exq;
  f3 := sqrt(exq)·sin(i/2)·cos(Ω); f4 := sqrt(exq)·sin(i/2)·sin(Ω);
  f5 := cos(Ω + ω + ν)/P;
  f6 := sin(Ω + ω + ν)/P;
  f7 := amp;
  f8 := ν0;
```

$$f1 := \frac{e\sim\cos(\omega\sim)}{\sqrt{-e\sim^{2}+1}}$$

$$f2 := \frac{e\sim\sin(\omega\sim)}{\sqrt{-e\sim^{2}+1}}$$

$$f3 := \left(-e\sim^{2}+1\right)^{1/4}\sin\left(\frac{1}{2}\,i\sim\right)\cos(\Omega\sim)$$

$$f4 := \left(-e\sim^{2}+1\right)^{1/4}\sin\left(\frac{1}{2}\,i\sim\right)\sin(\Omega\sim)$$

$$f5 := \frac{\cos(\Omega\sim+\omega\sim+\nu\sim)}{P}$$

$$f6 := \frac{\sin(\Omega\sim+\omega\sim+\nu\sim)}{P}$$

$$f7 := amp$$

$$f8 := \nu0 \tag{7.1}$$

```
> A := matrix([[diff(f1, e), diff(f1, ω), diff(f1, i), diff(f1, Ω), diff(f1, P),
      diff(f1, ν), diff(f1, amp), diff(f1, ν0)],
            [diff(f2, e), diff(f2, ω), diff(f2, i), diff(f2, Ω), diff(f2, P), diff(f2,
      ν), diff(f2, amp), diff(f2, ν0)],
            [diff(f3, e), diff(f3, ω), diff(f3, i), diff(f3, Ω), diff(f3, P), diff(f3,
      ν), diff(f3, amp), diff(f3, ν0)],
            [diff(f4, e), diff(f4, ω), diff(f4, i), diff(f4, Ω), diff(f4, P), diff(f4,
      ν), diff(f4, amp), diff(f4, ν0)],
```

[diff(f5, e), diff(f5, ω), diff(f5, i), diff(f5, Ω), diff(f5, P), diff(f5, ν), diff(f5, amp), diff(f5, ν0)],
[diff(f6, e), diff(f6, ω), diff(f6, i), diff(f6, Ω), diff(f6, P), diff(f6, ν), diff(f6, amp), diff(f6, ν0)],
[diff(f7, e), diff(f7, omega), diff(f7, i), diff(f7, Omega), diff(f7, P), diff(f7, ν), diff(f7, amp), diff(f7, ν0)], [diff(f8, e), diff(f8, omega), diff(f8, i), diff(f8, Omega), diff(f8, P), diff(f8, ν), diff(f8, amp), diff(f8, ν0)]])

;
:

$$A := \left[\left[\frac{\cos(\omega\sim)}{\sqrt{-e\sim^2+1}} + \frac{e\sim^2\cos(\omega\sim)}{(-e\sim^2+1)^{3/2}}, -\frac{e\sim\sin(\omega\sim)}{\sqrt{-e\sim^2+1}}, 0, 0, 0, 0, 0, 0\right], \tag{7.2}$$

$$\left[\frac{\sin(\omega\sim)}{\sqrt{-e\sim^2+1}} + \frac{e\sim^2\sin(\omega\sim)}{(-e\sim^2+1)^{3/2}}, \frac{e\sim\cos(\omega\sim)}{\sqrt{-e\sim^2+1}}, 0, 0, 0, 0, 0, 0\right],$$

$$\left[-\frac{1}{2}\frac{\sin\left(\frac{1}{2}i\sim\right)\cos(\Omega\sim)\,e\sim}{(-e\sim^2+1)^{3/4}}, 0, \frac{1}{2}(-e\sim^2+1)^{1/4}\cos\left(\frac{1}{2}i\sim\right)\cos(\Omega\sim),\right.$$

$$\left.-(-e\sim^2+1)^{1/4}\sin\left(\frac{1}{2}i\sim\right)\sin(\Omega\sim), 0, 0, 0, 0\right],$$

$$\left[-\frac{1}{2}\frac{\sin\left(\frac{1}{2}i\sim\right)\sin(\Omega\sim)\,e\sim}{(-e\sim^2+1)^{3/4}}, 0, \frac{1}{2}(-e\sim^2+1)^{1}\right.$$

$$\left.{}^{/4}\cos\left(\frac{1}{2}i\sim\right)\sin(\Omega\sim), (-e\sim^2+1)^{1/4}\sin\left(\frac{1}{2}i\sim\right)\cos(\Omega\sim), 0, 0, 0, 0\right],$$

$$\left[0, -\frac{\sin(\Omega\sim+\omega\sim+\nu\sim)}{P}, 0, -\frac{\sin(\Omega\sim+\omega\sim+\nu\sim)}{P},\right.$$

$$\left.-\frac{\cos(\Omega\sim+\omega\sim+\nu\sim)}{P^2}, -\frac{\sin(\Omega\sim+\omega\sim+\nu\sim)}{P}, 0, 0\right],$$

$$\left[0, \frac{\cos(\Omega\sim+\omega\sim+\nu\sim)}{P}, 0, \frac{\cos(\Omega\sim+\omega\sim+\nu\sim)}{P}, -\frac{\sin(\Omega\sim+\omega\sim+\nu\sim)}{P^2},\right.$$

$$\left.\frac{\cos(\Omega\sim+\omega\sim+\nu\sim)}{P}, 0, 0\right],$$

$$\left[0, 0, 0, 0, 0, 0, 1, 0\right],$$

$$\left[0, 0, 0, 0, 0, 0, 0, 1\right]\Big]$$

> $simplify(linalg[det](A)), factor\left(simplify\left(\frac{linalg[det](A)\cdot P\cdot amp}{2\cdot\sin\left(\frac{i}{2}\right)\cdot\cos\left(\frac{i}{2}\right)}\cdot dvdM\right)\right),$

$factor\left(simplify\left(\frac{linalg[det](A)\cdot P\cdot amp}{2\cdot\sin\left(\frac{i}{2}\right)\cdot\cos\left(\frac{i}{2}\right)}\cdot dvdMu\right)\right);$

On multiplie par P et a et on divise par sin(i) pour tenir compte des priors (ln P, amp^2, et sin(i)) sur P, amp et i
p(pars).d(pars)=C.d(pars)=C.J.d(elts)=C.J.d(e).d(omega).d(i).d(Omega).d(P).d(nu).d(amp)
=C.J.d(e).d(omega).d(Omega).d(cos(i)).d(lnP).d(lnamp).d(M).P.amp.(d(nu)/dM)/sin(i)

$$-\frac{1}{2}\,\frac{e\sim\cos\left(\frac{1}{2}\,i\sim\right)\sin\left(\frac{1}{2}\,i\sim\right)}{\left(-e\sim^2+1\right)^{3/2}P^3},\ \frac{1}{4}\,\frac{e\sim amp\,(1+e\sim\cos(\nu\sim))^2}{(e\sim-1)^3\,(e\sim+1)^3\,P^2},\ \frac{1}{4}\,\frac{amp\,e\sim}{(e\sim-1)\,(e\sim+1)\,P^2\,(e\sim\cos(u\sim)-1)^2} \tag{7.3}$$

>

>

> $h1 := \frac{e\cdot\cos(\omega)}{exq};\ h2 := \frac{e\cdot\sin(\omega)}{exq};$

$h3 := \sin\left(\frac{i}{2}\right)\cdot\cos(\Omega);\ h4 := \sin\left(\frac{i}{2}\right)\cdot\sin(\Omega);$

$h5 := \frac{1}{P}\cdot\cos(\text{omega}+\nu);$

$h6 := \frac{1}{P}\cdot\sin(\text{omega}+\nu);$

$$h1 := \frac{e\sim\cos(\omega\sim)}{\sqrt{-e\sim^2+1}}$$

$$h2 := \frac{e\sim\sin(\omega\sim)}{\sqrt{-e\sim^2+1}}$$

$$h3 := \sin\left(\frac{1}{2}\,i\sim\right)\cos(\Omega\sim)$$

$$h4 := \sin\left(\frac{1}{2}\, i\sim\right) \sin(\Omega\sim)$$

$$h5 := \frac{\cos(\omega\sim + \nu\sim)}{P}$$

$$h6 := \frac{\sin(\omega\sim + \nu\sim)}{P} \tag{8.1}$$

```
> B := matrix([[diff(h1, e), diff(h1, ω), diff(h1, i), diff(h1, Ω), diff(h1, P),
    diff(h1, ν)],
        [diff(h2, e), diff(h2, ω), diff(h2, i), diff(h2, Ω), diff(h2, P),
    diff(h2, ν)],
        [diff(h3, e), diff(h3, ω), diff(h3, i), diff(h3, Ω), diff(h3, P),
    diff(h3, ν)],
        [diff(h4, e), diff(h4, ω), diff(h4, i), diff(h4, Ω), diff(h4, P),
    diff(h4, ν)],
        [diff(h5, e), diff(h5, ω), diff(h5, i), diff(h5, Ω), diff(h5, P),
    diff(h5, ν)],
        [diff(h6, e), diff(h6, ω), diff(h6, i), diff(h6, Ω), diff(h6, P),
    diff(h6, ν)]]);
```

$$B := \left[\left[\frac{\cos(\omega\sim)}{\sqrt{-e\sim^2+1}} + \frac{e\sim^2 \cos(\omega\sim)}{(-e\sim^2+1)^{3/2}}, -\frac{e\sim \sin(\omega\sim)}{\sqrt{-e\sim^2+1}}, 0, 0, 0, 0\right],\right.$$
$$\left[\frac{\sin(\omega\sim)}{\sqrt{-e\sim^2+1}} + \frac{e\sim^2 \sin(\omega\sim)}{(-e\sim^2+1)^{3/2}}, \frac{e\sim \cos(\omega\sim)}{\sqrt{-e\sim^2+1}}, 0, 0, 0, 0\right],$$
$$\left[0, 0, \frac{1}{2}\cos\left(\frac{1}{2}\, i\sim\right)\cos(\Omega\sim), -\sin\left(\frac{1}{2}\, i\sim\right)\sin(\Omega\sim), 0, 0\right],$$
$$\left[0, 0, \frac{1}{2}\cos\left(\frac{1}{2}\, i\sim\right)\sin(\Omega\sim), \sin\left(\frac{1}{2}\, i\sim\right)\cos(\Omega\sim), 0, 0\right],$$
$$\left[0, -\frac{\sin(\omega\sim+\nu\sim)}{P}, 0, 0, -\frac{\cos(\omega\sim+\nu\sim)}{P^2}, -\frac{\sin(\omega\sim+\nu\sim)}{P}\right],$$
$$\left.\left[0, \frac{\cos(\omega\sim+\nu\sim)}{P}, 0, 0, -\frac{\sin(\omega\sim+\nu\sim)}{P^2}, \frac{\cos(\omega\sim+\nu\sim)}{P}\right]\right] \tag{8.2}$$

```
> simplify(linalg[det](B)), factor(simplify(linalg[det](B)·P / (2·sin(i/2)·cos(i/2)) ·dvdM));
```

$$-\frac{1}{2}\,\frac{e\sim \cos\left(\frac{1}{2}\, i\sim\right)\sin\left(\frac{1}{2}\, i\sim\right)}{(e\sim^2-1)^2 P^3}, -\frac{1}{4}\,\frac{e\sim (1+e\sim\cos(\nu\sim))^2}{P^2\,(-(e\sim-1)\,(e\sim+1))^{7/2}} \tag{8.3}$$

```
>
>
>
```

```
> A := matrix([[diff(g1, e), diff(g1, ω), diff(g1, i), diff(g1, Ω), diff(g1, P),
    diff(g1, ν)],
        [diff(g2, e), diff(g2, ω), diff(g2, i), diff(g2, Ω), diff(g2, P), diff(g2,
    ν)],
        [diff(g3, e), diff(g3, ω), diff(g3, i), diff(g3, Ω), diff(g3, P), diff(g3,
    ν)],
        [diff(g4, e), diff(g4, ω), diff(g4, i), diff(g4, Ω), diff(g4, P), diff(g4,
    ν)],
        [diff(g5, e), diff(g5, ω), diff(g5, i), diff(g5, Ω), diff(g5, P), diff(g5,
    ν)],
        [diff(g6, e), diff(g6, ω), diff(g6, i), diff(g6, Ω), diff(g6, P), diff(g6,
    ν)]]);
```

$$A := \left[\left[\frac{\cos(\Omega\sim+\omega\sim)}{\sqrt{-e\sim^2+1}}+\frac{e\sim^2\cos(\Omega\sim+\omega\sim)}{(-e\sim^2+1)^{3/2}}, -\frac{e\sim\sin(\Omega\sim+\omega\sim)}{\sqrt{-e\sim^2+1}}, 0, -\frac{e\sim\sin(\Omega\sim+\omega\sim)}{\sqrt{-e\sim^2+1}}, 0, 0\right],\right.$$

$$\left[\frac{\sin(\Omega\sim+\omega\sim)}{\sqrt{-e\sim^2+1}}+\frac{e\sim^2\sin(\Omega\sim+\omega\sim)}{(-e\sim^2+1)^{3/2}}, \frac{e\sim\cos(\Omega\sim+\omega\sim)}{\sqrt{-e\sim^2+1}}, 0, \frac{e\sim\cos(\Omega\sim+\omega\sim)}{\sqrt{-e\sim^2+1}}, 0, 0\right],$$

$$\left[\frac{e\sim^{\xi}\xi\sin\left(\frac{1}{2}i\sim\right)\cos(-\omega\sim+\Omega\sim)(-e\sim^2+1)^{1/4}}{e\sim}-\frac{1}{2}\frac{e\sim^{\xi}\sin\left(\frac{1}{2}i\sim\right)\cos(-\omega\sim+\Omega\sim)e\sim}{(-e\sim^2+1)^{3/4}}, e\sim^{\xi}(-e\sim^2+1)^{1/4}\sin\left(\frac{1}{2}i\sim\right)\sin(-\omega\sim+\Omega\sim), \frac{1}{2}e\sim^{\xi}\cos\left(\frac{1}{2}i\sim\right)\cos(-\omega\sim+\Omega\sim)(-e\sim^2+1)^{1/4}, -e\sim^{\xi}(-e\sim^2+1)^{1/4}\sin\left(\frac{1}{2}i\sim\right)\sin(-\omega\sim+\Omega\sim), 0, 0\right],$$

$$\left[-\frac{e\sim^{\xi}\xi(-e\sim^2+1)^{1/4}\sin\left(\frac{1}{2}i\sim\right)\sin(-\omega\sim+\Omega\sim)}{e\sim}+\frac{1}{2}\frac{e\sim^{\xi}\sin\left(\frac{1}{2}i\sim\right)\sin(-\omega\sim+\Omega\sim)e\sim}{(-e\sim^2+1)^{3/4}}, e\sim^{\xi}\sin\left(\frac{1}{2}i\sim\right)\cos(-\omega\sim+\Omega\sim)(-e\sim^2+1)^{1/4}, -\frac{1}{2}e\sim^{\xi}(-e\sim^2+1)^{1/4}\cos\left(\frac{1}{2}i\sim\right)\sin(-\omega\sim+\Omega\sim),\right.$$

(2)

$$-e\sim^{\xi}\sin\left(\frac{1}{2}\,i\sim\right)\cos(-\omega\sim+\Omega\sim)\,(-e\sim^{2}+1)^{1/4},0,0\Big],$$

$$\Big[0,-\frac{\sin(\Omega\sim+\omega\sim+\nu\sim)}{P},0,-\frac{\sin(\Omega\sim+\omega\sim+\nu\sim)}{P},-\frac{\cos(\Omega\sim+\omega\sim+\nu\sim)}{P^{2}},$$

$$-\frac{\sin(\Omega\sim+\omega\sim+\nu\sim)}{P}\Big],$$

$$\Big[0,\frac{\cos(\Omega\sim+\omega\sim+\nu\sim)}{P},0,\frac{\cos(\Omega\sim+\omega\sim+\nu\sim)}{P},-\frac{\sin(\Omega\sim+\omega\sim+\nu\sim)}{P^{2}},$$

$$\frac{\cos(\Omega\sim+\omega\sim+\nu\sim)}{P}\Big]\Big]$$

> $simplify(linalg[det](A)), factor\left(simplify\left(\frac{linalg[det](A)\cdot P}{2\cdot\sin\left(\frac{i}{2}\right)\cdot\cos\left(\frac{i}{2}\right)}\cdot dvdM\right)\right);$

$$\frac{e\sim^{1+2\xi}\cos\left(\frac{1}{2}\,i\sim\right)\sin\left(\frac{1}{2}\,i\sim\right)}{(-e\sim^{2}+1)^{3/2}P^{3}},\ -\frac{1}{2}\,\frac{e\sim^{1+2\xi}(1+e\sim\cos(\nu\sim))^{2}}{(e\sim-1)^{3}(e\sim+1)^{3}P^{2}} \tag{3}$$

> $h1 := \frac{e\cdot\cos(\omega)}{exq};\ h2 := \frac{e\cdot\sin(\omega)}{exq};$
$h3 := \sin\left(\frac{i}{2}\right)\cdot\cos(\Omega);\ h4 := \sin\left(\frac{i}{2}\right)\cdot\sin(\Omega);$
$h5 := \frac{1}{P}\cdot\cos(\mathrm{omega}+\nu);$
$h6 := \frac{1}{P}\cdot\sin(\mathrm{omega}+\nu);$

$$h1 := \frac{e\sim\cos(\omega\sim)}{\sqrt{-e\sim^{2}+1}}$$

$$h2 := \frac{e\sim\sin(\omega\sim)}{\sqrt{-e\sim^{2}+1}}$$

$$h3 := \sin\left(\frac{1}{2}\,i\sim\right)\cos(\Omega\sim)$$

$$h4 := \sin\left(\frac{1}{2}\,i\sim\right)\sin(\Omega\sim)$$

$$h5 := \frac{\cos(\omega\sim+\nu\sim)}{P}$$

$$h6 := \frac{\sin(\omega\sim+\nu\sim)}{P} \tag{4}$$

> $B := matrix([[diff(h1, e), diff(h1, \omega), diff(h1, i), diff(h1, \Omega), diff(h1, P), diff(h1, \nu)],$
$[diff(h2, e), diff(h2, \omega), diff(h2, i), diff(h2, \Omega), diff(h2, P), diff(h2,$

$v)$],
[$diff(h3, e)$, $diff(h3, \omega)$, $diff(h3, i)$, $diff(h3, \Omega)$, $diff(h3, P)$, $diff(h3, v)$],
[$diff(h4, e)$, $diff(h4, \omega)$, $diff(h4, i)$, $diff(h4, \Omega)$, $diff(h4, P)$, $diff(h4, v)$],
[$diff(h5, e)$, $diff(h5, \omega)$, $diff(h5, i)$, $diff(h5, \Omega)$, $diff(h5, P)$, $diff(h5, v)$],
[$diff(h6, e)$, $diff(h6, \omega)$, $diff(h6, i)$, $diff(h6, \Omega)$, $diff(h6, P)$, $diff(h6, v)$]]);

$$B := \left[\left[\frac{\cos(\omega\sim)}{\sqrt{-e\sim^2+1}} + \frac{e\sim^2\cos(\omega\sim)}{(-e\sim^2+1)^{3/2}}, -\frac{e\sim\sin(\omega\sim)}{\sqrt{-e\sim^2+1}}, 0, 0, 0, 0\right], \left[\frac{\sin(\omega\sim)}{\sqrt{-e\sim^2+1}} + \frac{e\sim^2\sin(\omega\sim)}{(-e\sim^2+1)^{3/2}}, \frac{e\sim\cos(\omega\sim)}{\sqrt{-e\sim^2+1}}, 0, 0, 0, 0\right], \left[0, 0, \frac{1}{2}\cos\left(\frac{1}{2}i\sim\right)\cos(\Omega\sim), -\sin\left(\frac{1}{2}i\sim\right)\sin(\Omega\sim), 0, 0\right], \left[0, 0, \frac{1}{2}\cos\left(\frac{1}{2}i\sim\right)\sin(\Omega\sim), \sin\left(\frac{1}{2}i\sim\right)\cos(\Omega\sim), 0, 0\right], \left[0, -\frac{\sin(\omega\sim+v\sim)}{P}, 0, 0, -\frac{\cos(\omega\sim+v\sim)}{P^2}, -\frac{\sin(\omega\sim+v\sim)}{P}\right], \left[0, \frac{\cos(\omega\sim+v\sim)}{P}, 0, 0, -\frac{\sin(\omega\sim+v\sim)}{P^2}, \frac{\cos(\omega\sim+v\sim)}{P}\right]\right] \quad (5)$$

> $simplify(linalg[det](B)), factor\left(simplify\left(\frac{linalg[det](B)\cdot P}{2\cdot\sin\left(\frac{i}{2}\right)\cdot\cos\left(\frac{i}{2}\right)}\cdot dvdM\right)\right);$

$$-\frac{1}{2}\frac{e\sim\cos\left(\frac{1}{2}i\sim\right)\sin\left(\frac{1}{2}i\sim\right)}{(e\sim^2-1)^2P^3}, -\frac{1}{4}\frac{e\sim(1+e\sim\cos(v\sim))^2}{P^2(-(e\sim-1)(e\sim+1))^{7/2}} \quad (6)$$

> $solve(x^2+y^2=e^2\cdot(1-e^2), e);$

Warning, solve may be ignoring assumptions on the input variables.

$$\frac{1}{2}\sqrt{2+2\sqrt{-4x^2-4y^2+1}}, -\frac{1}{2}\sqrt{2+2\sqrt{-4x^2-4y^2+1}}, \frac{1}{2}\sqrt{2-2\sqrt{-4x^2-4y^2+1}}, -\frac{1}{2}\sqrt{2-2\sqrt{-4x^2-4y^2+1}} \quad (7)$$

Cas 4 : Astrométrie + Vitesse radiale,

>

> $f1 := \frac{e \cdot \cos(\text{omega})}{exq}$; $f2 := \frac{e \cdot \sin(\text{omega})}{exq}$;

$f3 := \text{sqrt}(exq) \cdot \sin\left(\frac{i}{2}\right) \cdot \cos(\text{Omega})$; $f4 := \text{sqrt}(exq) \cdot \sin\left(\frac{i}{2}\right) \cdot \sin(\text{Omega})$;

$f5 := \frac{1}{P} \cdot \cos(\text{Omega} + \text{omega} + \nu)$;

$f6 := \frac{1}{P} \cdot \sin(\text{Omega} + \text{omega} + \nu)$;

$f7 := \nu 0$;

$f8 := \ln(a)$; $f9 := Amp$;

$$f1 := \frac{e\sim \cos(\omega\sim)}{\sqrt{-e\sim^2 + 1}}$$

$$f2 := \frac{e\sim \sin(\omega\sim)}{\sqrt{-e\sim^2 + 1}}$$

$$f3 := (-e\sim^2 + 1)^{1/4} \sin\left(\frac{1}{2}\, i\sim\right) \cos(\Omega\sim)$$

$$f4 := (-e\sim^2 + 1)^{1/4} \sin\left(\frac{1}{2}\, i\sim\right) \sin(\Omega\sim)$$

$$f5 := \frac{\cos(\Omega\sim + \omega\sim + \nu\sim)}{P}$$

$$f6 := \frac{\sin(\Omega\sim + \omega\sim + \nu\sim)}{P}$$

$$f7 := \nu 0$$

$$f8 := \ln(a\sim)$$

$$f9 := Amp \tag{8}$$

> $A := matrix([[diff(f1, e), diff(f1, \omega), diff(f1, i), diff(f1, \Omega), diff(f1, P), diff(f1, \nu), diff(f1, \nu 0), diff(f1, a), diff(f1, Amp)],$
$[diff(f2, e), diff(f2, \omega), diff(f2, i), diff(f2, \Omega), diff(f2, P), diff(f2, \nu), diff(f2, \nu 0), diff(f2, a), diff(f2, Amp)],$
$[diff(f3, e), diff(f3, \omega), diff(f3, i), diff(f3, \Omega), diff(f3, P), diff(f3, \nu), diff(f3, \nu 0), diff(f3, a), diff(f3, Amp)],$
$[diff(f4, e), diff(f4, \omega), diff(f4, i), diff(f4, \Omega), diff(f4, P), diff(f4, \nu), diff(f4, \nu 0), diff(f4, a), diff(f4, Amp)],$
$[diff(f5, e), diff(f5, \omega), diff(f5, i), diff(f5, \Omega), diff(f5, P), diff(f5, \nu), diff(f5, \nu 0), diff(f5, a), diff(f5, Amp)],$
$[diff(f6, e), diff(f6, \omega), diff(f6, i), diff(f6, \Omega), diff(f6, P), diff(f6, \nu), diff(f6, \nu 0), diff(f6, a), diff(f6, Amp)],$
$[diff(f7, e), diff(f7, \text{omega}), diff(f7, i), diff(f7, \text{Omega}), diff(f7, P), diff(f7, \nu), diff(f7, \nu 0), diff(f7, a), diff(f7, Amp)],$
$[diff(f8, e), diff(f8, \text{omega}), diff(f8, i), diff(f8, \text{Omega}), diff(f8, P), diff(f8, \nu), diff(f8, \nu 0), diff(f8, a), diff(f8, Amp)], [diff(f9, e), diff(f9, \text{omega}), diff(f9, i), diff(f9, \text{Omega}), diff(f9, P), diff(f9, \nu), diff(f9, \nu 0), diff(f9, a), diff(f9, Amp)]]);$

$$A := \left[\left[\frac{\cos(\omega\sim)}{\sqrt{-e\sim^2+1}} + \frac{e\sim^2\cos(\omega\sim)}{(-e\sim^2+1)^{3/2}}, -\frac{e\sim\sin(\omega\sim)}{\sqrt{-e\sim^2+1}}, 0, 0, 0, 0, 0, 0, 0\right],\right. \tag{9}$$

$$\left[\frac{\sin(\omega\sim)}{\sqrt{-e\sim^2+1}} + \frac{e\sim^2\sin(\omega\sim)}{(-e\sim^2+1)^{3/2}}, \frac{e\sim\cos(\omega\sim)}{\sqrt{-e\sim^2+1}}, 0, 0, 0, 0, 0, 0, 0\right],$$

$$\left[-\frac{1}{2}\frac{\sin\left(\frac{1}{2}i\sim\right)\cos(\Omega\sim)\,e\sim}{(-e\sim^2+1)^{3/4}}, 0, \frac{1}{2}(-e\sim^2+1)^{1/4}\cos\left(\frac{1}{2}i\sim\right)\cos(\Omega\sim),\right.$$

$$\left.-(-e\sim^2+1)^{1/4}\sin\left(\frac{1}{2}i\sim\right)\sin(\Omega\sim), 0, 0, 0, 0, 0\right],$$

$$\left[-\frac{1}{2}\frac{\sin\left(\frac{1}{2}i\sim\right)\sin(\Omega\sim)\,e\sim}{(-e\sim^2+1)^{3/4}}, 0, \frac{1}{2}(-e\sim^2+1)^{1/4}\cos\left(\frac{1}{2}i\sim\right)\sin(\Omega\sim), (\right.$$

$$\left.-e\sim^2+1)^{1/4}\sin\left(\frac{1}{2}i\sim\right)\cos(\Omega\sim), 0, 0, 0, 0, 0\right],$$

$$\left[0, -\frac{\sin(\Omega\sim+\omega\sim+\nu\sim)}{P}, 0, -\frac{\sin(\Omega\sim+\omega\sim+\nu\sim)}{P}, -\frac{\cos(\Omega\sim+\omega\sim+\nu\sim)}{P^2},\right.$$

$$\left.-\frac{\sin(\Omega\sim+\omega\sim+\nu\sim)}{P}, 0, 0, 0\right],$$

$$\left[0, \frac{\cos(\Omega\sim+\omega\sim+\nu\sim)}{P}, 0, \frac{\cos(\Omega\sim+\omega\sim+\nu\sim)}{P}, -\frac{\sin(\Omega\sim+\omega\sim+\nu\sim)}{P^2},\right.$$

$$\left.\frac{\cos(\Omega\sim+\omega\sim+\nu\sim)}{P}, 0, 0, 0\right],$$

$$\left[0, 0, 0, 0, 0, 0, 1, 0, 0\right],$$

$$\left[0, 0, 0, 0, 0, 0, 0, \frac{1}{a\sim}, 0\right],$$

$$\left.\left[0, 0, 0, 0, 0, 0, 0, 0, 1\right]\right]$$

```
> simplify(linalg[det](A)), factor(simplify(linalg[det](A)·P·a·Amp / (2·sin(i/2)·cos(i/2)) · dvdM));
```

$$factor\left(simplify\left(\frac{linalg[det](A)\cdot P\cdot a\cdot Amp}{2\cdot\sin\left(\frac{i}{2}\right)\cdot\cos\left(\frac{i}{2}\right)}\cdot dvdMu\right)\right)$$

On multiplie par P, a et on divise par cos(i) pour tenir compte des priors (ln P et sin (i), ln a)
sur a, P et i (voir plus haut)

$$-\frac{1}{2}\,\frac{e\sim\cos\left(\frac{1}{2}\,i\sim\right)\sin\left(\frac{1}{2}\,i\sim\right)}{\left(-e\sim^2+1\right)^{3/2}P^3\,a\sim},\ \frac{1}{4}\,\frac{e\sim Amp\,(1+e\sim\cos(v\sim))^2}{(e\sim-1)^3\,(e\sim+1)^3\,P^2}$$
$$\frac{1}{4}\,\frac{Amp\,e\sim}{(e\sim-1)\,(e\sim+1)\,P^2\,(e\sim\cos(u\sim)-1)^2} \quad \textbf{(10)}$$

Cas 8 : Vitesses radiales seules

> $g1 := \frac{2\cdot Pi}{P}\cdot\cos(v);\ g2 := \frac{2\cdot Pi}{P}\cdot\sin(v);\ g3 := \frac{e\cdot\cos(\omega)}{exq};\ g4 := \frac{e\cdot\sin(omega)}{exq};$
$g5 := Amp;\ g6 := v0;$

$$g1 := \frac{2\,\pi\cos(v\sim)}{P}$$
$$g2 := \frac{2\,\pi\sin(v\sim)}{P}$$
$$g3 := \frac{e\sim\cos(\omega\sim)}{\sqrt{-e\sim^2+1}}$$
$$g4 := \frac{e\sim\sin(\omega\sim)}{\sqrt{-e\sim^2+1}}$$
$$g5 := Amp$$
$$g6 := v0 \quad \textbf{(10.1)}$$

> $A := matrix([[diff(g1, e), diff(g1, \omega), diff(g1, v0), diff(g1, Amp), diff(g1, P), diff(g1, v)],$
$[diff(g2, e), diff(g2, \omega), diff(g2, v0), diff(g2, Amp), diff(g2, P), diff(g2, v)],$
$[diff(g3, e), diff(g3, \omega), diff(g3, v0), diff(g3, Amp), diff(g3, P), diff(g3, v)],$
$[diff(g4, e), diff(g4, \omega), diff(g4, v0), diff(g4, Amp), diff(g4, P), diff(g4, v)],$
$[diff(g5, e), diff(g5, \omega), diff(g5, v0), diff(g5, Amp), diff(g5, P), diff(g5, v)],$
$[diff(g6, e), diff(g6, \omega), diff(g6, v0), diff(g6, Amp), diff(g6, P), diff(g6, v)]]);$

$$A := \left[\left[0, 0, 0, 0, -\frac{2\,\pi\cos(v\sim)}{P^2}, -\frac{2\,\pi\sin(v\sim)}{P}\right],\right. \quad \textbf{(10.2)}$$

$$\left[0, 0, 0, 0, -\frac{2\,\pi \sin(\nu\sim)}{P^2}, \frac{2\,\pi \cos(\nu\sim)}{P}\right],$$
$$\left[\frac{\cos(\omega\sim)}{\sqrt{-e\sim^2+1}} + \frac{e\sim^2 \cos(\omega\sim)}{(-e\sim^2+1)^{3/2}}, -\frac{e\sim \sin(\omega\sim)}{\sqrt{-e\sim^2+1}}, 0, 0, 0, 0\right],$$
$$\left[\frac{\sin(\omega\sim)}{\sqrt{-e\sim^2+1}} + \frac{e\sim^2 \sin(\omega\sim)}{(-e\sim^2+1)^{3/2}}, \frac{e\sim \cos(\omega\sim)}{\sqrt{-e\sim^2+1}}, 0, 0, 0, 0\right],$$
$$\left[0, 0, 0, 1, 0, 0\right],$$
$$\left[0, 0, 1, 0, 0, 0\right]\Big]$$

```
> simplify(linalg[det](A)), factor(simplify(linalg[det](A)·P·Amp·dvdM));
```

$$\frac{4\,e\sim \pi^2}{(e\sim^2-1)^2\,P^3}, \frac{4\,e\sim \pi^2\,Amp\,(1+e\sim \cos(\nu\sim))^2}{P^2\,(-(e\sim-1)\,(e\sim+1))^{7/2}} \tag{10.3}$$

```
> -147.5 + 180;
```

$$32.5 \tag{10.4}$$

```
>
```

```
> dvdM;
```

$$\frac{(1+e\sim \cos(\nu\sim))^2}{(-e\sim^2+1)^{3/2}} \tag{11}$$

Cas 6 : Astrométrie, variables universelles, masse contrainte

```
> g1 := q·cos(Ω + ω); g2 := q·sin(Ω + ω);
  g3 := sin(i/2)·cos(omega-Ω); g4 := sin(i/2)·sin(omega-Ω);
  g5 := e;
  g6 := s;
```

$$g1 := q\cos(\Omega\sim+\omega\sim)$$
$$g2 := q\sin(\Omega\sim+\omega\sim)$$
$$g3 := \sin\left(\frac{1}{2}\,i\sim\right)\cos(-\omega\sim+\Omega\sim)$$
$$g4 := -\sin\left(\frac{1}{2}\,i\sim\right)\sin(-\omega\sim+\Omega\sim)$$
$$g5 := e\sim$$
$$g6 := s \tag{11.1}$$

```
> A := matrix([[diff(g1, e), diff(g1, ω), diff(g1, i), diff(g1, Ω), diff(g1, q),
      diff(g1, s)],
            [diff(g2, e), diff(g2, ω), diff(g2, i), diff(g2, Ω), diff(g2, q),
      diff(g2, s)],
            [diff(g3, e), diff(g3, ω), diff(g3, i), diff(g3, Ω), diff(g3, q),
```

$diff(g3, s)],$
$[diff(g4, e), diff(g4, \omega), diff(g4, i), diff(g4, \Omega), diff(g4, q),$
$diff(g4, s)],$
$[diff(g5, e), diff(g5, \omega), diff(g5, i), diff(g5, \Omega), diff(g5, q),$
$diff(g5, s)],$
$[diff(g6, e), diff(g6, \omega), diff(g6, i), diff(g6, \Omega), diff(g6, q),$
$diff(g6, s)]]);$

$$A := \left[\left[0, -q\sin(\Omega\sim + \omega\sim), 0, -q\sin(\Omega\sim + \omega\sim), \cos(\Omega\sim + \omega\sim), 0\right], \left[0, q\cos(\Omega\sim + \omega\sim), 0, q\cos(\Omega\sim + \omega\sim), \sin(\Omega\sim + \omega\sim), 0\right], \left[0, \sin\left(\frac{1}{2} i\sim\right)\sin(-\omega\sim + \Omega\sim), \frac{1}{2}\cos\left(\frac{1}{2} i\sim\right)\cos(-\omega\sim + \Omega\sim), -\sin\left(\frac{1}{2} i\sim\right)\sin(-\omega\sim + \Omega\sim), 0, 0\right], \left[0, \sin\left(\frac{1}{2} i\sim\right)\cos(-\omega\sim + \Omega\sim), -\frac{1}{2}\cos\left(\frac{1}{2} i\sim\right)\sin(-\omega\sim + \Omega\sim), -\sin\left(\frac{1}{2} i\sim\right)\cos(-\omega\sim + \Omega\sim), 0, 0\right], \left[1, 0, 0, 0, 0, 0\right], \left[0, 0, 0, 0, 0, 1\right]\right] \quad (11.2)$$

> $dvdMq := \frac{(1 + e\cdot\cos(\nu))^2}{(1+e)^{\frac{3}{2}}};$

$$dvdMq := \frac{(1 + e\sim\cos(\nu\sim))^2}{(e\sim + 1)^{3/2}} \quad (11.3)$$

> $simplify(linalg[det](A)), factor\left(simplify\left(\frac{linalg[det](A)\cdot q}{2\cdot\sin\left(\frac{i}{2}\right)\cdot\cos\left(\frac{i}{2}\right)}\cdot dsdt\right)\right);$

$$\cos\left(\frac{1}{2} i\sim\right)\sin\left(\frac{1}{2} i\sim\right) q, \frac{1}{2} q^2\, dsdt \quad (11.4)$$

> $dvdM;$

$$\frac{(1 + e\sim\cos(\nu\sim))^2}{(-e\sim^2 + 1)^{3/2}} \quad (11.5)$$

>

On multiplie par Pq et on divise par sin(i) pour tenir compte
des priors proportionnels à (ln Pq et sin(i) et M) sur Pq,et i (Pq^2*q^3=GM)
p(pars).d(pars)=C.d(pars)=C.J.d(elts)=C.J.d(e).d(omega).d(i).d(Omega).d(Pq).d(nu).d(a)
=C.J.d(e).d(omega).d(Omega).d(cos(i)).d(lnPq).d(Mq).Pq.(d(nu)/d(Mq))/sin(i)

> $dsdt;$

$$dsdt \tag{11.6}$$

Cas 6b : Vitesses radiales, variables universelles

```
> g1 := q·cos(ω); g2 := q·sin(ω);
  g3 := ln(m);
  g4 := e;
  g5 := s;
```

$$g1 := q\cos(\omega\sim)$$
$$g2 := q\sin(\omega\sim)$$
$$g3 := \ln(m)$$
$$g4 := e\sim$$
$$g5 := s \tag{11.1.1}$$

```
> A := matrix([[diff(g1, e), diff(g1, ω), diff(g1, m), diff(g1, q), diff(g1, s)],
              [diff(g2, e), diff(g2, ω), diff(g2, m), diff(g2, q), diff(g2, s)],
              [diff(g3, e), diff(g3, ω), diff(g3, m), diff(g3, q), diff(g3, s)],
              [diff(g4, e), diff(g4, ω), diff(g4, m), diff(g4, q), diff(g4, s)],
              [diff(g5, e), diff(g5, ω), diff(g5, m), diff(g5, q), diff(g5, s)]]);
```

$$A := \begin{bmatrix} 0 & -q\sin(\omega\sim) & 0 & \cos(\omega\sim) & 0 \\ 0 & q\cos(\omega\sim) & 0 & \sin(\omega\sim) & 0 \\ 0 & 0 & \frac{1}{m} & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 \end{bmatrix} \tag{11.1.2}$$

```
> dvdMq := (1 + e·cos(ν))^2 / (1 + e)^(3/2);
```

$$dvdMq := \frac{(1+e\sim\cos(\nu\sim))^2}{(e\sim+1)^{3/2}} \tag{11.1.3}$$

```
> simplify(linalg[det](A)), factor(simplify(linalg[det](A)·q·m·dsdt));
```

$$-\frac{q}{m},\ -q^2\ dsdt \tag{11.1.4}$$

```
> dvdM;
```

$$\frac{(1+e\sim\cos(\nu\sim))^2}{(-e\sim^2+1)^{3/2}} \tag{11.1.5}$$

```
>
```

On multiplie par m, Pq et on divise par sin(i) pour tenir compte
des priors proportionnels à (lm(m) et ln Pq et sin(i) et M) sur Pq,et i (Pq^2*
q^3=GM)
p(pars).d(pars)=C.d(pars)=C.J.d(elts)=C.J.d(e).d(omega).d(Pq).d(nu).d(a)
=C.J.d(e).d(omega).d(Omega).d(cos(i)).d(lnPq).d(Mq).Pq.(d(nu)/d
(Mq))/sin(i)

> $dsdt;$

$$dsdt \tag{11.1.6}$$

Cas 7 : Astrometrie, variables universelles, rotation variables

> $\text{alpha} := \frac{\mu}{q} \cdot (1 - e);\ C := \text{sqrt}(q \cdot \text{mu} \cdot (1 + e));$
$g1 := q^{\text{xi}} \cdot \cos(i);\ g2 := \frac{q^{\text{xi}} \cdot \cos(\text{omega}) \cdot \sin(\text{omega}) \cdot \sin(i)^2}{\text{sqrt}(1 - \sin(i)^2 \cdot \sin(\text{omega})^2)};$
$g3 := -\cos(\text{omega}) \cdot \sin(\text{Omega}) - \sin(\text{omega}) \cdot \cos(i) \cdot \cos(\text{Omega});$
$g4 := \cos(\text{omega}) \cdot \cos(\text{Omega}) - \cos(i) \cdot \sin(\text{omega}) \cdot \sin(\text{Omega});$
$g5 := e;$
$g6 := s;$

$$\alpha := \frac{\mu\,(-e\sim + 1)}{q}$$
$$C := \sqrt{q\,\mu\,(e\sim + 1)}$$
$$g1 := q^{\xi} \cos(i\sim)$$
$$g2 := \frac{q^{\xi} \cos(\omega\sim)\,\sin(\omega\sim)\,\sin(i\sim)^2}{\sqrt{1 - \sin(i\sim)^2 \sin(\omega\sim)^2}}$$
$$g3 := -\cos(\omega\sim)\,\sin(\Omega\sim) - \cos(\Omega\sim)\,\cos(i\sim)\,\sin(\omega\sim)$$
$$g4 := -\cos(i\sim)\,\sin(\Omega\sim)\,\sin(\omega\sim) + \cos(\Omega\sim)\,\cos(\omega\sim)$$
$$g5 := e\sim$$
$$g6 := s \tag{12.1}$$

> $\text{alpha};\ C;$

$$\frac{\mu\,(-e\sim + 1)}{q}$$
$$\sqrt{q\,\mu\,(e\sim + 1)} \tag{12.2}$$

> $A := \text{matrix}([[\text{diff}(g1, e), \text{diff}(g1, \omega), \text{diff}(g1, i), \text{diff}(g1, \Omega), \text{diff}(g1, q), \text{diff}(g1, s)],$
$[\text{diff}(g2, e), \text{diff}(g2, \omega), \text{diff}(g2, i), \text{diff}(g2, \Omega), \text{diff}(g2, q), \text{diff}(g2, s)],$
$[\text{diff}(g3, e), \text{diff}(g3, \omega), \text{diff}(g3, i), \text{diff}(g3, \Omega), \text{diff}(g3, q), \text{diff}(g3, s)],$
$[\text{diff}(g4, e), \text{diff}(g4, \omega), \text{diff}(g4, i), \text{diff}(g4, \Omega), \text{diff}(g4, q), \text{diff}(g4, s)],$
$[\text{diff}(g5, e), \text{diff}(g5, \omega), \text{diff}(g5, i), \text{diff}(g5, \Omega), \text{diff}(g5, q), \text{diff}(g5, s)],$
$[\text{diff}(g6, e), \text{diff}(g6, \omega), \text{diff}(g6, i), \text{diff}(g6, \Omega), \text{diff}(g6, q), \text{diff}(g6, s)]]);$

(12.3)

$$A := \left[\left[0, 0, -q^{\xi}\sin(i\sim), 0, \frac{q^{\xi}\,\xi\cos(i\sim)}{q}, 0\right],\right.$$

$$\left[0, -\frac{q^{\xi}\sin(\omega\sim)^2\sin(i\sim)^2}{\sqrt{1-\sin(i\sim)^2\sin(\omega\sim)^2}} + \frac{q^{\xi}\cos(\omega\sim)^2\sin(i\sim)^2}{\sqrt{1-\sin(i\sim)^2\sin(\omega\sim)^2}}\right.$$

$$+\frac{q^{\xi}\cos(\omega\sim)^2\sin(\omega\sim)^2\sin(i\sim)^4}{\left(1-\sin(i\sim)^2\sin(\omega\sim)^2\right)^{3/2}}, \frac{2\,q^{\xi}\cos(\omega\sim)\,\sin(\omega\sim)\,\sin(i\sim)\,\cos(i\sim)}{\sqrt{1-\sin(i\sim)^2\sin(\omega\sim)^2}}$$

$$+\frac{q^{\xi}\cos(\omega\sim)\,\sin(\omega\sim)^3\sin(i\sim)^3\cos(i\sim)}{\left(1-\sin(i\sim)^2\sin(\omega\sim)^2\right)^{3/2}}, 0,$$

$$\left.\frac{q^{\xi}\,\xi\cos(\omega\sim)\,\sin(\omega\sim)\,\sin(i\sim)^2}{q\sqrt{1-\sin(i\sim)^2\sin(\omega\sim)^2}}, 0\right],$$

$$\left[0, \sin(\Omega\sim)\,\sin(\omega\sim) - \cos(\Omega\sim)\,\cos(i\sim)\,\cos(\omega\sim),\right.$$

$$\cos(\Omega\sim)\,\sin(i\sim)\,\sin(\omega\sim), -\cos(\Omega\sim)\,\cos(\omega\sim)$$

$$+\cos(i\sim)\,\sin(\Omega\sim)\,\sin(\omega\sim), 0, 0],$$

$$\left[0, -\cos(i\sim)\,\cos(\omega\sim)\,\sin(\Omega\sim) - \cos(\Omega\sim)\,\sin(\omega\sim),\right.$$

$$\sin(i\sim)\,\sin(\Omega\sim)\,\sin(\omega\sim), -\cos(\omega\sim)\,\sin(\Omega\sim)$$

$$-\cos(\Omega\sim)\,\cos(i\sim)\,\sin(\omega\sim), 0, 0],$$

$$\left[1, 0, 0, 0, 0, 0\right],$$

$$\left.\left[0, 0, 0, 0, 0, 1\right]\right] \tag{12.3}$$

> $dvdMq := \frac{(1+e\cdot\cos(\nu))^2}{(1+e)^{\frac{3}{2}}};$

$$dvdMq := \frac{(1+e\sim\cos(\nu\sim))^2}{(e\sim+1)^{3/2}} \tag{12.4}$$

> $simplify(linalg[det](A)); jaco := simplify\left(\frac{linalg[det](A)\cdot q}{\sin(i)}\cdot dsdt\right);$

$$-\xi\,q^{2\xi-1}\sqrt{\left(1-\cos(\omega\sim)^2\right)\cos(i\sim)^2+\cos(\omega\sim)^2}\,\sin(\omega\sim)^2\sin(i\sim)^3$$

$$jaco := -dsdt\,\xi\,q^{2\xi}\sqrt{\left(1-\cos(\omega\sim)^2\right)\cos(i\sim)^2+\cos(\omega\sim)^2}\,\sin(\omega\sim)^2\sin(i\sim)^2 \tag{12.5}$$

On multiplie par q et on divise par sin(i) pour tenir compte
des priors proportionnels à (ln q et sin(i) et M) sur q,et i (Pq^2*q^3=GM)
p(pars).d(pars)=C.d(pars)=C.J.d(elts)=C.J.d(e).d(omega).d(i).d(Omega).d(q).d(s).d(a)
=C.J.d(e).d(omega).d(Omega).d(cos(i)).d(lnq).d(t).q.(d(s)/d(t))/sin(i)

> $jacos := simplify(factor(jaco), \{\cos(i)^2 = 1 - \sin(i)^2, \cos(omega)^2 = 1 - \sin(omega)^2\});$

$$jacos := -dsdt\,\xi\,(q^{\xi})^2 \sqrt{1 - \sin(i\sim)^2 \sin(\omega\sim)^2}\, \sin(\omega\sim)^2 \sin(i\sim)^2 \quad (12.6)$$

> $jacos := factor(jacos);$

$$jacos := -dsdt\,\xi\,(q^{\xi})^2 \sqrt{-(\sin(\omega\sim)\sin(i\sim) - 1)(\sin(\omega\sim)\sin(i\sim) + 1)}\, \sin(\omega\sim)^2 \sin(i\sim)^2 \quad (12.7)$$

> $factor(simplify(simplify(jacos), \{\cos(omega)^2 = 1 - \sin(omega)^2, \cos(i)^2 = 1 - \sin(i)^2\}));$

$$-dsdt\,\xi\,(q^{\xi})^2 \sqrt{-(\sin(\omega\sim)\sin(i\sim) - 1)(\sin(\omega\sim)\sin(i\sim) + 1)}\, \sin(\omega\sim)^2 \sin(i\sim)^2 \quad (12.8)$$

> $factor(g1^2 + g2^2);$

$$-\left((q^{\xi})^2 (\cos(\omega\sim)^2 \sin(\omega\sim)^2 \sin(i\sim)^4 - \sin(\omega\sim)^2 \sin(i\sim)^2 \cos(i\sim)^2 + \cos(i\sim)^2)\right) \Big/ \left((\sin(\omega\sim)\sin(i\sim) - 1)(\sin(\omega\sim)\sin(i\sim) + 1)\right) \quad (12.9)$$

>

Cas 10 : Astrométrie, variables universelles

> $g1 := q;\ g2 := e\cdot\cos(\Omega + \omega);\ g3 := e\cdot\sin(\Omega + \omega);$
$g4 := \sin\left(\frac{i}{2}\right)\cdot\cos(omega\text{-}\Omega);\ g5 := \sin\left(\frac{i}{2}\right)\cdot\sin(omega\text{-}\Omega);$
$g6 := P;\ g7 := s\cdot f(e);$

$$g1 := q$$
$$g2 := e\sim\cos(\Omega\sim + \omega\sim)$$
$$g3 := e\sim\sin(\Omega\sim + \omega\sim)$$
$$g4 := \sin\left(\frac{1}{2}\, i\sim\right)\cos(-\omega\sim + \Omega\sim)$$
$$g5 := -\sin\left(\frac{1}{2}\, i\sim\right)\sin(-\omega\sim + \Omega\sim)$$
$$g6 := P$$
$$g7 := s\, f(e\sim) \quad (12.1.1)$$

> $A := matrix([[diff(g1, e), diff(g1, \omega), diff(g1, i), diff(g1, \Omega), diff(g1, q), diff(g1, s), diff(g1, P)],$
$[diff(g2, e), diff(g2, \omega), diff(g2, i), diff(g2, \Omega), diff(g2, q), diff(g2, s), diff(g2, P)],$
$[diff(g3, e), diff(g3, \omega), diff(g3, i), diff(g3, \Omega), diff(g3, q), diff(g3, s), diff(g3, P)],$
$[diff(g4, e), diff(g4, \omega), diff(g4, i), diff(g4, \Omega), diff(g4, q), diff(g4, s), diff(g4, P)],$
$[diff(g5, e), diff(g5, \omega), diff(g5, i), diff(g5, \Omega), diff(g5, q), diff(g5, s), diff(g5, P)],$

$[diff(g6, e), diff(g6, \omega), diff(g6, i), diff(g6, \Omega), diff(g6, q), diff(g6, s), diff(g6, P)],$
$[diff(g7, e), diff(g7, \omega), diff(g7, i), diff(g7, \Omega), diff(g7, q), diff(g7, s), diff(g7, P)]]);$

;

$$A := \left[\left[0, 0, 0, 0, 1, 0, 0\right], \left[\cos(\Omega\sim + \omega\sim), -e\sim \sin(\Omega\sim + \omega\sim), 0, -e\sim \sin(\Omega\sim + \omega\sim), 0, 0, 0\right], \left[\sin(\Omega\sim + \omega\sim), e\sim \cos(\Omega\sim + \omega\sim), 0, e\sim \cos(\Omega\sim + \omega\sim), 0, 0, 0\right], \left[0, \sin\left(\frac{1}{2} i\sim\right) \sin(-\omega\sim + \Omega\sim), \frac{1}{2} \cos\left(\frac{1}{2} i\sim\right) \cos(-\omega\sim + \Omega\sim), -\sin\left(\frac{1}{2} i\sim\right) \sin(-\omega\sim + \Omega\sim), 0, 0, 0\right], \left[0, \sin\left(\frac{1}{2} i\sim\right) \cos(-\omega\sim + \Omega\sim), -\frac{1}{2} \cos\left(\frac{1}{2} i\sim\right) \sin(-\omega\sim + \Omega\sim), -\sin\left(\frac{1}{2} i\sim\right) \cos(-\omega\sim + \Omega\sim), 0, 0, 0\right], \left[0, 0, 0, 0, 0, 0, 1\right], \left[s\left(\frac{d}{de\sim} f(e\sim)\right), 0, 0, 0, 0, f(e\sim), 0\right]\right] \quad \textbf{(12.1.2)}$$

> $dvdMq := \frac{(1 + e\cdot\cos(\nu))^2}{(1 + e)^{\frac{3}{2}}};$

$$dvdMq := \frac{(1 + e\sim \cos(\nu\sim))^2}{(e\sim + 1)^{3/2}} \quad \textbf{(12.1.3)}$$

> $simplify(linalg[det](A)), factor\left(simplify\left(\frac{linalg[det](A)\cdot q\cdot P}{2\cdot\sin\left(\frac{i}{2}\right)\cdot\cos\left(\frac{i}{2}\right)}\cdot dsdt\right)\right);$

$$\cos\left(\frac{1}{2} i\sim\right) \sin\left(\frac{1}{2} i\sim\right) f(e\sim)\, e\sim, \frac{1}{2} q\, P\, dsdt\, f(e\sim)\, e\sim \quad \textbf{(12.1.4)}$$

> $dvdM;$

$$\frac{(1 + e\sim \cos(\nu\sim))^2}{(-e\sim^2 + 1)^{3/2}} \quad \textbf{(12.1.5)}$$

>

On multiplie par P et q et on divise par sin(i) pour tenir compte
des priors proportionnels à (ln P, ln q et sin(i)) sur P, q,et i (P^2*q^3=GM)
p(pars).d(pars)=C.d(pars)=C.J.d(elts)=C.J.d(e).d(omega).d(i).d(Omega).d(Pq).d(nu).d(a)
=C.J.d(e).d(omega).d(Omega).d(cos(i)).d(lnP).d(q).P.(d(nu)/d(Mq))/sin(i)

> $dsdt;$

$$dsdt \tag{12.1.6}$$

Cas 6b : Vitesses radiales, variables universelles

> $g1 := q\cdot\cos(\omega);\ g2 := q\cdot\sin(\omega);$
$g3 := \ln(m);$
$g4 := e;$
$g5 := s;$

$$g1 := q\cos(\omega\sim)$$
$$g2 := q\sin(\omega\sim)$$
$$g3 := \ln(m)$$
$$g4 := e\sim$$
$$g5 := s \tag{12.1.1.1}$$

> $A := matrix([[diff(g1, e), diff(g1, \omega), diff(g1, m), diff(g1, q), diff(g1, s)],$
$[diff(g2, e), diff(g2, \omega), diff(g2, m), diff(g2, q), diff(g2, s)],$
$[diff(g3, e), diff(g3, \omega), diff(g3, m), diff(g3, q), diff(g3, s)],$
$[diff(g4, e), diff(g4, \omega), diff(g4, m), diff(g4, q), diff(g4, s)],$
$[diff(g5, e), diff(g5, \omega), diff(g5, m), diff(g5, q), diff(g5, s)]])$
;

$$A := \begin{bmatrix} 0 & -q\sin(\omega\sim) & 0 & \cos(\omega\sim) & 0 \\ 0 & q\cos(\omega\sim) & 0 & \sin(\omega\sim) & 0 \\ 0 & 0 & \frac{1}{m} & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 \end{bmatrix} \tag{12.1.1.2}$$

> $dvdMq := \frac{(1+e\cdot\cos(\nu))^2}{(1+e)^{\frac{3}{2}}};$

$$dvdMq := \frac{(1+e\sim\cos(\nu\sim))^2}{(e\sim+1)^{3/2}} \tag{12.1.1.3}$$

> $simplify(linalg[det](A)),\ factor(simplify(linalg[det](A)\cdot q\cdot m\cdot dsdt));$

$$-\frac{q}{m},\ -q^2\ dsdt \tag{12.1.1.4}$$

> $dvdM;$

$$\frac{(1+e\sim\cos(\nu\sim))^2}{(-e\sim^2+1)^{3/2}} \tag{12.1.1.5}$$

>

On multiplie par m, Pq et on divise par sin(i) pour tenir compte
des priors proportionnels à (lm(m) et ln Pq et sin(i) et M) sur Pq,et i (Pq^2*
q^3=GM)
p(pars).d(pars)=C.d(pars)=C.J.d(elts)=C.J.d(e).d(omega).d(Pq).d(nu).d(a)

=C.J.d(e).d(omega).d(Omega).d(cos(i)).d(lnPq).d(Mq).Pq.(d(nu)/d(Mq))/sin(i)

```
> dsdt;
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$$dsdt \tag{12.1.1.6}$$

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>
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